

SURFACE GRAFTING ONTO POLYPROPYLENE — A SURVEY OF RECENT DEVELOPMENTS*

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1. INTRODUCTION

A graft copolymer is a polymer which is comprised of molecules with one or more species of blocks connected as side chains to the backbone, having constitutional or configurational features different from those in the main chain.

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The simplest case of a graft copolymer can be represented by (I):



where a sequence of the M monomer unit is the backbone, a sequence of the G-unit is the side chain (graft) and X is the unit in the backbone to which the graft is attached. In graft copolymers, the backbone and side chains may be homopolymeric or copolymeric units.

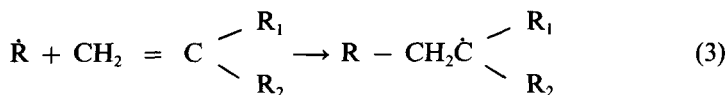
2. METHODS OF GRAFTING

Formation of graft copolymers usually involves diffusion across a phase boundary between a monomer and the polymeric material. Graft copolymers can also be prepared by copolymerizing preformed branches (containing a polymerizable endgroup) with the monomer which will form the main chain. The branch must have an end group capable of copolymerizing with the monomer by the method of polymerization being used. Preformed branches can be also coupled on to a preformed main chain.¹

2.1. Free-radical mechanism

The free-radical polymerization methods are the oldest and most widely used procedures for the synthesis of graft polymers because they are relatively simple. There are five methods of grafting a polymer by free-radical mechanism.

2.1.1. Chemical method — The free-radicals which are the sites of grafting are obtained by initiators such as benzoyl peroxide (BPO) or azobisisobutyronitrile (AIBN) in this method (see Ref. 24). The synthesis of sequential polymers is illustrated in the following sequence:



where RH is the polymer. The generated radical adds to the monomer to form the homopolymer chain in eq. (3). The radical may attach to an in-chain double

In the pre-irradiation technique, free radicals are trapped if the polymer is irradiated in the absence of air and afterwards monomer is introduced into the system to graft onto the polymer backbone. Sakurada *et al.*² have grafted polystyrene onto cellulose fibers by this technique in dry air. They confirmed that grafting occurred by trapped radicals and not by peroxide groups.

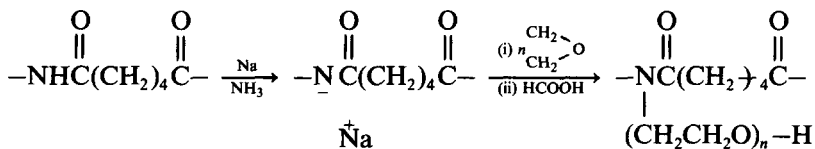
2.1.4. Plasma grafting — The grafting of plastics as well as naturally occurring fibers by glow discharge is known as plasma grafting.² The exposure of polymers to glow discharge produces free radicals² on the polymer backbone, which subsequently add monomer. A low temperature discharge is a complex system consisting of electrons, atoms, ionized species and excited atoms and molecules. These particles cause radiation effects on the surface of the material as well as in the interior of the substance. Millard *et al.*³ have grafted fluorocarbon “monomers” (i.e. hexafluoroethane and tetrafluoroethylene) onto a variety of surfaces (i.e. polymers, wood and textile) by exposing them to glow discharge.

2.1.5. Mechanochemical grafting — The successive mechanical and ultrasonic shearing forces may also cause polymer degradation, usually by a free-radical mechanism. The mechanical degradation may be carried out by mastication, milling, extrusion, high speed stirring or shaking, forcing the polymer solution through an orifice, or freezing and thawing of polymer solutions. The thus generated free-radicals, in the presence of monomers, initiate the grafting of the latter onto the polymer surface.

2.2. Ionic mechanism

Ionic polymerization is a good technique for the preparation of graft copolymers.

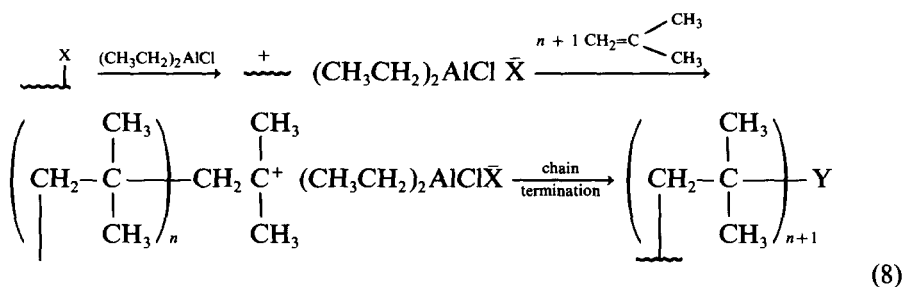
2.2.1. Anionic method — Since the discovery of ‘living’ polymerization⁴ anionic polymerization has become an excellent method for synthesis of block and graft copolymers. Graft copolymerization is initiated by anions generated by the reaction of bases with acidic protons in the main polymer chain. For example, polyamides^{5,6} are metalated with sodium in liquid ammonia and then grafted with a monomeric unit:



(7)

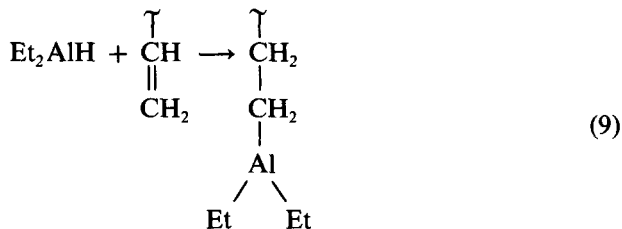
2.2.2. Cationic method — The initiation reactions between labile alkyl halides

and Lewis acids have been utilized for cationic grafting onto halogenated polymers⁷ as follows



2.3. Coordination mechanism

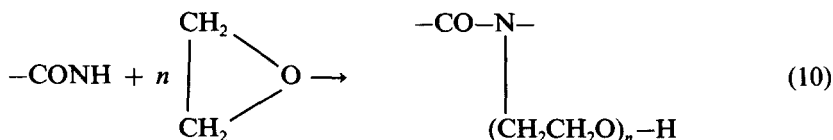
Stereospecific initiators can give stereoblock copolymers which contain isotactic and heterotactic sequences. Greber⁸ grafted olefins onto poly(styrene-Co-butadiene) using a Ziegler-Natta coordination initiator system where diethylaluminium hydride reacts with pendant groups to form a macromolecular trialkylaluminium:



Trialkylaluminium units with TiCl_4 or TiCl_3 form a Ziegler-Natta initiator system which is used in the grafting of α -olefins onto the preformed macromolecular trialkylaluminium. By this technique, poly(styrene-Co-butadiene)-g-ethylene or isotactic polypropylene was obtained. Extraction freed the polymer of initiator residues.

2.4. Coupling mechanism

Polymers containing active hydrogen can be used for the synthesis of graft copolymers. Poly(ethylene oxide) is easily grafted onto nylon since the hydrogen atom of the amide nitrogen is readily removed:



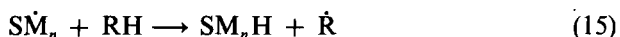
Graft copolymers can also be prepared by coupling two or more polymers containing suitable functional groups.

3. KINETICS OF SURFACE GRAFTING

In the first set of preceding methods of graft copolymerization, radicals are the initiating species for grafting. These radicals are formed by a chemical initiator in initiator graft copolymerization, by a high-energy radiation in radiation-induced graft copolymerization and by light radiation in photo-graft copolymerization. The free radicals formed on a polymeric chain act as a site of attack for the monomer to be grafted.

3.1. *Kinetics of grafting by free radicals*

The scheme of the initiation and propagation reactions is as follows:



where $S-S$ is the initiator and RH is a macromolecule. The primary radicals from reactions 11–13 add to the first monomer unit. A hydrogen atom of the main polymeric chain is abstracted either by a primary radical $\dot{S}$ (reaction 14) or by a growing molecule (reaction 15). Finally, reaction 16 represents the addition of the monomer to the active centre and formation of a graft copolymer.²⁴

The grafting reaction of vinyl monomer on polypropylene preferentially occurs on the tertiary site:



The reaction rate is higher when hydrogen is bound to a tertiary carbon, irrespective of the type of attacking radical $\dot{X}$. Formation of the active centre $\dot{R}$ on the polymer backbone is the step preceding grafting.

The initiation is followed by a series of propagation and termination reactions:



By single coupling (reaction 18), only one macromolecule P is obtained while by

disproportionation, two macromolecules are obtained. Grafted macroradicals may terminate by chain transfer



where T is a chain transfer agent and may be homopolymer, graft copolymer, monomer or the solvent.

The rate of grafting reaction r_i of the system is

$$r_i = -d[M]_i/dt = k_p[\dot{R}][M] \quad (20)$$

where $\dot{R}$ is the concentration of the grafted growing macroradicals. The rate of termination reaction r_{tr} is given by

$$r_{tr} = k_{t[\dot{R}][\dot{Y}]} + k_{t[\dot{R}][\dot{T}]} = [\dot{R}] \{k_{t[\dot{Y}]} + k_{t[\dot{T}]}\}. \quad (21)$$

3.2. Kinetics of grafting in the presence of peroxide

Hydroperoxide groups are formed on the polypropylene backbone by irradiation of polypropylene film in air. The thermal decomposition of hydroperoxide⁸ (POOH) groups will produce alkoxy ($\dot{P}O$) and hydroxyl ($\dot{O}H$) radicals. The decomposition kinetic scheme is



and the rate of decomposition can be written:

$$R_d = -d(POOH)/dt = k_d(POOH). \quad (23)$$

The equation (23) on integration leads to

$$(POOH)_t = (POOH)_0 e^{-k_d t} \quad (24)$$

where $(POOH)_0$ is the polypropylene hydroperoxide concentration decomposed at zero time and $(POOH)_t$ decomposed during t time, k_d is the rate constant of decomposition. The rearrangement of eqs (23) and (24) leads to

$$R_d = k_d(POOH)_0 e^{-k_d t}. \quad (25)$$

The kinetic scheme of the radicals which initiate the grafting reaction is:

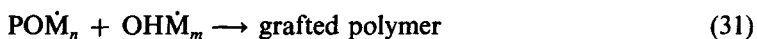
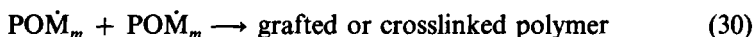
Initiation;



Propagation;



Termination;



but the rate of grafting is regarded as the rate of propagation in eq. (29).

$$R_g = k_p(\text{POM})(M) \quad (33)$$

where POM is the concentration of peroxide radicals. At the steady-state, the rate of termination is equal to the rate of hydroperoxide decomposition (radical production).

$$k_d(\text{POOH})_0 e^{-k_d t} = k'_t(\text{POM})^2 + k''_t(\text{POM})(\text{OHM}) + k'''_t(\text{OHM})^2 \quad (34)$$

A more simplified equation may be derived if the concentration of the OHM radical equals that of the POM radical:

$$k_d(\text{POOH})_0 e^{-k_d t} = k_t(\text{POM})^2. \quad (35)$$

Rearranging eqs (34) and (35) yields:

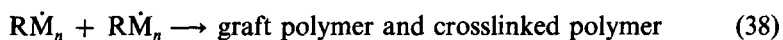
$$R_g = \frac{dG}{dt} = k_p(M)(k_d)^{1/2}(\text{POOH})_0^{1/2}(k_t)^{-1/2}e^{-k_d t/2}. \quad (36)$$

Integration of eq. (36) leads to

$$G = 2k_p(\text{POOH})_0^{1/2}(M)(k_t k_d)^{-1/2}(1 - e^{-k_d/2}). \quad (37)$$

3.3. Kinetics of grafting by trapped radicals

The graft copolymerization reaction can also be initiated by trapped radicals formed by irradiation in the absence of air. The number of trapped radicals and growing chains decreases with time due to chain termination and radical recombination.



The rate of termination can be written

$$R_t = -d(\text{RM})/dt = k_t(\text{RM})^2 \quad (39)$$

where (RM) is the concentration of growing radicals. On integration it becomes

$$\text{RM} = (\text{RM})_0[1 + k_t(\text{RM})_0 t] - 1 \quad (40)$$

where (RM)₀ is the concentration of trapped radical at zero time. The rate of grafting can be written:

$$R_g = dG/dt = k_p(M)(\text{RM}). \quad (41)$$

On integration eqs (40) and (41) lead to

$$G = k_p k_t^{-1} (M) \ln [1 + k_t (R\dot{M})_0 t]. \quad (42)$$

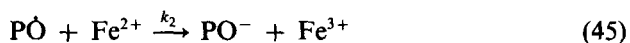
If the reaction is carried out for sufficient time t , then eq. (42) can be simplified:

$$G = k_p k_t^{-1} (M) \ln [k_t (R\dot{M})_0 t]. \quad (43)$$

It is clearly indicated by eq. (43) that the degree of grafting should be directly proportional to monomer concentration (M), logarithm of reaction time t and trapped radical concentration. But the concentration of trapped radical is directly proportional to irradiation dose at low doses. Therefore, the degree of grafting should be proportional to the logarithm of irradiation dose while other parameters are constant.

3.4. Kinetics of grafting in the presence of metal ion catalyst

The grafting is increased by using reducing agents ($\text{FeSO}_4/\text{CuCl}_2$) which decompose the previously peroxidized species by irradiation in air, thereby converting hydroxyl radical ($\dot{\text{O}}\text{H}$) to hydroxide anion⁹ (OH^-):



(radical scavenging)



(chain initiation)

where M is a monomer. Both the reactions (45) and (46) can proceed simultaneously but with different rates:

$$R_1 = k_1 [\text{P}\dot{\text{O}}][\text{Fe}^{2+}] \quad (47)$$

$$R_2 = k_2 [\text{P}\dot{\text{O}}][\text{M}]. \quad (48)$$

At a definite concentration of Fe^{2+} ion, the rate of radical scavenging reaches the rate of initiation, where

$$k_1 [\text{P}\dot{\text{O}}][\text{Fe}^{2+}] = k_2 [\text{P}\dot{\text{O}}][\text{M}] \quad (49)$$

or

$$k_1 [\text{Fe}^{2+}] = k_2 [\text{M}] \quad (50)$$

or

$$[\text{Fe}^{2+}] = \frac{k_2}{k_1} [\text{M}]. \quad (51)$$

Equation (51) confirms that a higher concentration of Fe^{2+} is required for a higher degree of grafting.

4. SURFACE GRAFTING OF POLYPROPYLENE

Compared to the large amount of research devoted towards developing new polymers and modifying the bulk properties of polymers by grafting,^{10,11} relatively little work has been done on the surface chemistry and physics of polymers. The surface modifying techniques presently in use are surface coating, surface degradation, surface hydrolysis, plasma discharge treatment, and radiation-induced, photo-induced or catalytic-initiated graft copolymerization. Modification of polymers through graft copolymerization offers an effective means for introducing some desirable properties into the polymer without affecting the architecture of the polymer backbone.

Polypropylene (PP) has an important place among the synthetic polymers because of its growing commercial applications. But the drawbacks of polypropylene (sensitive to photo-oxidation, difficult to dye, hydrophobic surface, etc.) restrict its use in several technologically important fields. These drawbacks can be overcome by graft polymerization onto the PP surface. By this technique the polymer acquires some additional properties needed for a particular application with no change to the original desirable properties. The removal of tertiary hydrogen atom from the PP backbone by graft copolymerization with a variety of monomers provides a powerful method for partial modification of PP and improvement of its properties.

4.1. Initiator (free-radical generator) grafting

Burri¹² grafted styrene onto a polypropylene surface in benzene solution at 85°C containing initiator ($\text{Bz}_2\text{O}_2/\text{AIBN}$) in the absence of air to avoid peroxide formation. A graft of styrene onto PP with good dyeability is prepared by mixing 90 parts peroxidized PP with 10 parts styrene. Above the T_m of PP, this material undergoes radical grafting.¹³ The grafting process of methyl methacrylate onto a PP surface was carried out through polymer hydroperoxide formation and through a chain transfer reaction.¹⁴ The hydroperoxide induced grafting proceeding by direct free-radical initiation at the hydroperoxide groups and via hydrogen atom transfer from PP to the PMMA radical. The extent of vinyl acetate grafting onto atactic and isotactic polypropylene depends on the temperature of grafting and not on the temperature of PP oxidation. Vinyl acetate was grafted onto PP¹⁵ by oxidizing the latter at 90°C for 24 hr. The vinyl acetate-g-PP showed better mechanical, thermal and dyeability properties²⁰ than the PP homopolymer. Grafting in this case was carried out by free-radical initiators at 60–110°C. The graft copolymerization of acrylonitrile upon PP fiber was carried out without previous activation of the polymer by boiling the latter in a saturated aqueous solution of acrylonitrile in air.¹⁶ Ferrous sulphate suppressed acrylonitrile homopolymerization and the presence of antioxidant in the mixture reduced or even completely inhibited the graft copolymerization.

Polystyrene, poly(methyl methacrylate) and poly(diethyl methylene malonate), the last two of which degrade by an unzipping mechanism,¹⁷ were grafted onto PP¹⁸ and the mixture was degraded at 324°C. Infrared analysis proved that the modified materials contained the respective monomer units on the PP surface, and were unchanged by successive fractionation. Such means that it is not a simple mixture of homopolymers. Vinyl pyridine, acrylamide and vinyl caprolactam were grafted onto PP fiber¹⁹ by swelling the fiber in toluene containing 10 wt% Bz₂O₂ at 70°C. The tensile strength of the fiber was unchanged after grafting. To graft vinyl pyrrolidone onto *i*-PP, the latter was peroxidized at 130°C in xylene and the solution was saturated with nitrogen.²⁰ Upon stirring of 10 parts vinylpyrrolidone solution with PP for 3 hr at 130°C, grafting occurred which was terminated by cold water. The grafted fiber had excellent dyeability.

Pavunec and coworkers²¹ have graft-copolymerized atactic PP with styrene and/or methyl methacrylate using benzoyl peroxide or dicumyl peroxide. The grafting efficiency of atactic PP is lower with methyl methacrylate than with styrene. They have observed a solvent effect on the rate of grafting and have found that the extent of grafting of methyl methacrylate was 20% more in benzene than in *n*-heptane. The increase of grafting efficiency in the good solvent was attributed to the higher solubility of poly(methyl methacrylate) macro-radicals which were maintained in a more expanded form, facilitating the complete utilization of their higher reactivity in the substitution reaction. But a change in solvent did not affect the grafting efficiency of styrene, which meant that polystyrene radicals played a less significant role in the initiation of grafting than those of poly(methyl methacrylate).

Divinylbenzene can be grafted upon PP fibers²² in toluene using benzoyl peroxide as a free-radical initiator. The degree of crosslinking of the modified fiber increases with the concentration of grafted divinylbenzene. Vinylpyridine is also grafted onto PP fiber by peroxidizing crystalline PP in an aqueous solution containing a non-ionic detergent by generating radicals²³ with Bz₂O₂. This process increased the dyeability of the fiber. The polymerization of methyl acrylate in an atactic PP solution in the presence of a radical initiator (benzoyl peroxide, *tert*-butyl peroxyvalate, α,α' -azodiisobutyronitrile, lauroyl peroxide) yields a mixture of grafted methyl acrylate-polypropylene copolymer and poly(methyl acrylate), which were separated by fractionation.²⁴

Nakatsuka *et al.*²⁵ have stabilized PP by grafting it with glycidyl methacrylate. Then, traces of 4-(3-(3,5-di-*tert*-butyl-4-hydroxyphenyl)propionamido)phenol were milled with the polypropylene-glycidyl methacrylate graft. Upon spinning a yarn at 270°C from this mixture, it was found to be stable for 54 hr at 140°C while unstabilized material lasted for 2 hr only. PP can also be made thermally stable²⁶ by grafting hydroperoxidized PP with glycidyl methacrylate and milling it with 0.3 parts of dodecylthiopropionic acid and 3-(3,5-di-*tert*-butyl-4-hydroxyphenyl)propionic acid at 220°C for 20 min. The spun yarn of this graft

was stable for 122 hr at 140°C. The crack-resistant ethylene-propylene graft (15 wt% PP) was prepared²⁷ by subjecting PP to oxidation in air and then grafting it in benzene at ~100°C under 1050 kg/cm² pressure of ethylene in presence of a free-radical initiator. PP was grafted at high temperature²⁸ with butyl methacrylate, methacrylic acid or ethylene dimethacrylate in the presence of dibenzyltin oxide. The 12% grafted PP has 250% greater impact strength than the ungrafted material increased dyeability.

Vinyl pyridine was grafted²⁹ onto peroxidized PP under a nitrogen atmosphere. The grafted fiber, having high dye affinity, was prepared by treating it with phosphite derivatives. Tris(nonylphenyl)phosphite/trilauryl thiophosphite (or similar additives) imparted high light fastness to the fiber. Kozima *et al.*³⁰ have also stabilized PP grafts (under γ -ray irradiation) by 2-methyl-5-vinyl pyridine (7 wt%) by treating the melt spun fiber of the graft polymer with 1 wt% each of Santarox and Cyabsorb UV-531 light stabilizers. The stabilized fiber had better dyeability, light fastness, weather resistance and thermal stability than the unstabilized fiber.

An increased rate of methyl methacrylate grafting onto PP was attained by swelling the latter in *n*-heptane and treating it with cumene hydroperoxide at 60°C. The cumene hydroperoxide generates free radicals which are necessary for radical grafting.³¹ PP grafted with maleic anhydride and styrene was reinforced with glass fiber which had been treated with (γ -aminopropyl) trimethoxysilane.³² The grafting of maleic anhydride onto PP was carried out³³ at 120°C in the presence of Bz₂O₂. This graft gave good bond strength to tinplate joints. The test pieces after injection molding at 220°C gave improved mechanical and thermal properties. Butadiene, vinyl chloride and isoprene were grafted in the gaseous phase onto a peroxidized PP surface.³⁴ The peroxidation of *i*-PP was carried out by treating it with H₂O₂ in air at 10 atm. for 35 hr at 65°C.

PP pellets were grafted with vinyl aromatic monomers in aqueous media and impregnated with blowing agents to give expandable beads.³⁵ The initiators of grafting used here were Bz₂O₂/dicumyl peroxide at 85°C. The grafting of unsaturated carboxylic acids ($\leq 50\%$) on PP in the presence of a radical initiator by melt mixing method improved its reinforceability.³⁶ The mechanical properties and heat resistance of glass reinforced PP are further markedly improved by grafting the PP with a small amount of acrylic acid or maleic anhydride. The reinforcing effect of glass fiber is affected by the adhesion between grafted PP and the glass fiber, the uniformity ratio of the glass fiber dispersion and the crystal structure of the grafted PP. Grafted PP exhibited good adhesion to glass fiber coupled with aminosilane or epoxysilane to improve the mechanical strength and heat resistance. Silanes, having both alkenyl and hydrolyzable groups were grafted onto PP film by kneading in the presence of radical initiator and antioxidant.³⁷ Benzoyl peroxide was used as an initiator and pentaerythritol tetrakis[3-(3,5-di-*tert*-butyl-4-hydroxyphenyl)propionate] as an antioxidant.

Graft polymers with good solubility, useful as emulsifiers, were prepared by

TABLE 1. Extent of grafting of ethyl acrylate onto atactic PP at $80 \pm 2^\circ\text{C}$ with various amounts of benzoyl peroxide (based on ethyl acrylate) at different reaction times

Benzoyl peroxide wt%	Conversion %				
	1 hr	2 hr	3 hr	4 hr	5 hr
0.08	1.73	3.85	17.71	23.85	41.17
0.1	23.08	41.91	53.09	58.48	58.48
0.2	27.70	49.24	63.08	72.32	76.16
0.4	32.32	52.32	63.85	69.24	69.98
0.6	40.00	60.76	69.24	75.40	78.48
0.8	46.16	64.97	72.32	77.68	80.77
1.0	52.32	69.24	76.94	81.62	84.62

polymerization of gaseous olefins and maleic anhydride onto a PP surface in chlorinated hydrocarbons containing radical initiators³⁸ at 80°C . Isotactic PP was extruded with 0.5% Bz_2O_2 and 3% vinyltrimethoxysilane at 210°C to give a graft polymer which was moisture curable and had excellent heat distortion resistance.³⁹ The graft was further modified by blending it with dibutyltin dilaurate. The Bz_2O_2 -initiated copolymerization of vinyl monomers at $60\text{--}80^\circ\text{C}$ on atactic PP gave graft copolymers with 20% grafting.⁴⁰

The solubility and film forming properties of PP were improved by grafting. A vinyl chloride-PP graft was prepared⁴¹ by grafting vinyl chloride at 96% saturated vapor pressure on PP powder containing an oil soluble initiator (peroxide solution in dioctyl phthalate). The radical grafting of atactic PP with maleic anhydride and styrene resulted⁴² in a degree of grafting up to 22%, but incompatibility of the two phases was observed by dynamic mechanical measurement. Guhaniyogi *et al.*⁴³ have grafted ethyl acrylate onto an atactic PP surface at 80°C in xylene containing benzoyl peroxide in an inert atmosphere. The effect of benzoyl peroxide (BPO) concentration is shown in Table 1. The rate of grafting increased with increasing BPO concentration at times up to 5 hr and the grafting occurred by a radical chain-transfer process. Homogeneous graft polymers were prepared⁴⁴ by dispersing atactic PP in vinyl acetate/methyl methacrylate mixtures at 25°C above the atactic PP T_g in the presence of a radical initiator. For this purpose equimolar ratios of atactic PP and vinyl acetate were dispersed in heptane and the mixture containing 0.2% Bz_2O_2 was heated for 6 hr at 60°C . Madhavan and Sharma⁴⁵ improved the surface properties of atactic PP by grafting it with ethyl acrylate in the presence of Bz_2O_2 in xylene.

Alkyl methacrylate-PP graft copolymers were prepared⁴⁶ using *tert*-butyl peroxybenzoate as an initiator. The laminates are good adhesives. Hope and Malley⁴⁷ synthesized olefinic polymer blend adhesives for bonding PP to polar substrates. These adhesives comprise PP grafted with unsaturated carboxylic

acids or their derivatives, and low density polyethylene. The maleic anhydride-PP graft copolymer and PP were blended in a mixer at 180°C to get the required adhesion. The rigidity, adhesion and printability of PP were modified by grafting vinyl monomers onto PP. The grafting was performed by suspending PP void particles with vinyl monomers in aqueous media.⁴⁸ The graft copolymer was kneaded with 2,6-di-*tert*-butyl-*p*-cresol and calcium stearate and pelletized and injection moulded to give a test piece with flexural modulus 1.72 GPa and tensile strength 35 MPa while unmodified PP particles have values of 1.37 GPa and 33 MPa, respectively. Polymers useful in adhesives, coatings, films, etc. were prepared by gas phase grafting of vinyl monomers onto porous PP particles in the presence of dicumyl peroxide at 140°C. To enhance the graft level in grafting of maleic anhydride onto PP surface, toluene was used as an interfacial agent.⁴⁹ Toluene enhanced the graft level significantly at higher initiator concentrations (12.5%). The addition of interfacial agents resulted in agglomeration in the final product due to uneven distribution of these agents in the reactor.

4.2. Radiation-induced grafting

Acrylonitrile was grafted onto a preirradiated PP surface in vacuum via a hydroperoxide decomposition mechanism.⁵⁰ In highly irradiated PP, however, diffusion of the monomer becomes the rate-limiting factor. To graft styrene onto the surface of *i*-PP, a film of the latter was immersed in styrene monomer and irradiated⁵¹ in a nitrogen atmosphere at 30°C with γ -rays from a source at a dose rate of 58 Gy/hr. At a lower dose rate, the grafting efficiency was independent of the film thickness, whereas a higher dose rate decreased the grafting efficiency. The PP fiber was modified by γ -ray radiolysis in the presence of acrylic acid or methyl methacrylate⁵² and the rate of grafting was found proportional to the dose and monomer concentration. The rate of radiation-induced grafting of styrene/methyl methacrylate mixtures onto PP was accelerated^{53,54} when monomer was diluted with methanol. This acceleration is due to the Trommsdorff effect of the solvent and swelling of the polymer. The swelling causes enhancement of monomer access to the grafting site. The Trommsdorff effect is brought about by the insolubility of the growing graft polystyrene. The effect of methanol on grafting of styrene and methyl methacrylate upon PP is shown in Fig. 1 where styrene grafting undergoes a sharp peak with decreasing methanol concentration, but methanol seems to have only a diluent effect in methyl acrylate grafting.

Acrylonitrile was grafted upon PP by addition of the monomer to a γ -irradiated PP film.⁵⁵ The grafting ratio was found proportional to the trapped free radical concentration. It decreased with increasing additional time between irradiation and monomer exposure. For grafting of chloromethylstyrene to PP fiber, the fiber was immersed in chloromethylstyrene at 60°C for deep swelling and then irradiated with γ -rays in a nitrogen atmosphere.⁵⁶ This yielded uniform grafting

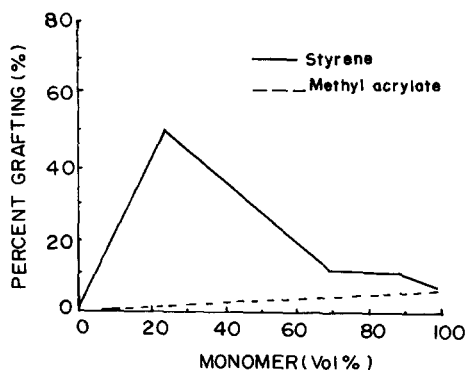


FIG. 1. Effect of methanol on the monomer grafting onto PP surface.

throughout the fiber and improved not only the dyeability but also the sticking temperature and moisture regain, thereby improving the antistatic properties. To graft acrylic acid on a PP surface, a PP film was γ -irradiated in an aqueous solution of acrylic acid⁵⁷ in an oxygen-free atmosphere at room temperature. To inhibit homopolymerization a small amount of ferrous sulphate was added to the solution.

Unezawa and Hirota⁵⁷ have grafted maleic anhydride onto pre-irradiated PP. The moisture absorption capacity and dyeability of PP fiber after grafting each of the following monomers was studied: acrylic acid, methyl methacrylate and vinyl acetate.⁵⁸ Peroxide groups and macro-radicals were formed on the fiber by γ -irradiation. The amount of grafted material increased with monomer concentration and temperature for a given irradiation time. Kimura and Kitano⁵⁸ observed an increase in grafting efficiency of styrene onto pre-irradiated PP in the presence of a reducing agent. The styrene or acrylonitrile was grafted⁵⁹ onto PP with α -ray irradiation by applying H_3BO_3 onto PP surface. The grafting was increased in the presence of oxygen because of peroxide radical formation (sites for grafting). The addition of an organic acid increases the grafting rate and reduces homopolymerization. If the same system was irradiated with neutrons instead of γ -rays⁶⁰ the effect of irradiation on the grafting rate was found to be much higher. A wide variety of vinyl monomers have been grafted⁶¹ onto PP surfaces using irradiation methods. The degree of grafting obtainable also depends upon the surface-to-volume ratio of the PP.

The investigations of Chapiro and others^{50,52,62,63} suggested that the method of pre-irradiation with γ -rays can be applied effectively to graft different monomers upon PP fibers and films. The penetration rate of grafting into the substrate depends upon the monomer diffusion and the concentration of free-radicals inside the PP sample. Armstrong and Walsh⁶⁴ grafted PP fibers with ethyl acrylate, vinylacetate and acrylonitrile in the vapor phase by mutual γ -irradiation. The modified fibers dyed well and moisture regaining capacity also

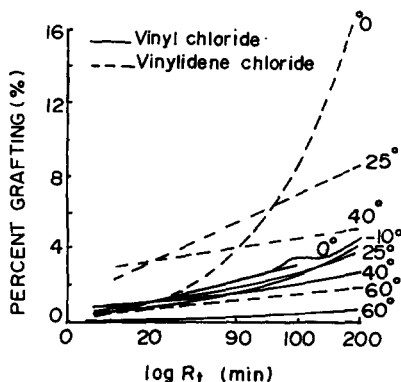


FIG. 2. Grafting rate of vinyl chloride and vinylidene chloride onto PP as a function of log of reaction time (R_t).

increased. Both radicals and peroxide groups produced by γ -irradiation in air were observed to initiate graft copolymerization⁶⁵ of olefinic monomers on PP. The grafting was apparently decreased if the irradiated polymer was stored for a long time even at 100°C. The 20% grafted PP film may be acid dyed.

Kamase and Hayakawa⁶⁶ have grafted alkyl methacrylates onto PP fiber upon exposure to γ -irradiation by two different techniques. The first is liquid phase grafting where the PP fiber was immersed into the liquid monomer. Butyl and propyl methacrylates were grafted to a very high extent, but the grafting efficiency of *iso*-butyl and *iso*-propyl methacrylates was found to be very low, probably due to the branched nature of the latter. Second is "gas phase" grafting where the polypropylene was suspended in the vapor of the monomer. The grafting rate was found to vary linearly with the vapor pressure of the monomer. The results showed that the rate of grafting at a constant monomer vapor pressure, was much less for vapor-exposed PP than for liquid-immersed PP. The reason may be a swelling effect in the liquid phase.

The graft copolymerization of gaseous vinyl chloride or vinylidene chloride upon γ -ray pre-irradiated PP fibers was carried out at -78°C in air.⁶⁷ The reactive peroxide radicals at the grafting site are active in grafting at room temperature also. Figure 2 shows the observed percentages of grafting versus logarithm of reaction time. Here the extent of grafting increases almost linearly with the logarithm of the reaction time but decreases with increasing temperature. The deviation from linearity in the initial stage is probably due to incomplete sorption of the monomer. The overall aspect of the reaction seemed to be similar for both monomer-polymer combinations. The grafting of vinylidene chloride, however, is restricted to a narrower temperature range because the vapor pressure of vinylidene chloride is lower at a given temperature.

The grafting of acrylonitrile upon preirradiated PP fiber was improved⁶⁸ by

dipping the fiber into water and then exposing it to acrylonitrile vapor at $\sim 75^{\circ}\text{C}$. Munari *et al.*⁶⁹ have observed a solvent effect on radiation-induced grafting of polyvinyl-pyridine onto PP film. The bulk grafting of PP occurs when the 4-vinylpyridine concentration exceeds 50 wt% in ethanol-4-vinylpyridine system because swelling of polymer becomes sufficient only at higher 4-vinylpyridine concentrations. In the PP-4-vinylpyridine-benzene system benzene swells the polymeric matrix better than does the ethanol-based system. Polyvinylpyridine, however, is not soluble in benzene. Therefore, increasing the amount of benzene retards the grafting process.

Odor⁷⁰ proved by X-ray diffraction studies that the grafting process of vinyl monomers upon PP fibers takes place in the amorphous phase at monomer contents up to 50%. Destruction of the crystal structure occurred only at monomer contents greater than 75%. The initial grafting is affected by the amount, orientation and segmental mobility of the amorphous phase. The grafting of gaseous monomers (vinylidene chloride and vinyl chloride) upon PP fiber was carried out by simultaneous γ -irradiation.⁷¹ The grafting rate was proportional to the vapor pressure of the monomer for both monomers, and to the 0.73 power of the radiation dose rate for the first and to the 1.1 power of the dose rate for the second monomer.

PP was grafted with a vinylpyridine (15%)-acrylate ester (5%) mixture after pre-irradiation at $< 5^{\circ}\text{C}$ by 15 kGy of electron beam radiation in an oxygen atmosphere.⁷² The grafted polymer so formed had superior dyeability, printability, oil resistance and antistatic properties to unmodified PP. Improved dyeability, light fastness and tensile strength of PP fiber was achieved by grafting onto its surface methacrylic acid, acrylamide or their mixture by γ -irradiation.⁷³ Neill⁷⁴ has grafted acrylic acid onto a 70 micrometer thick *i*-PP film previously peroxidized by 400 keV (64 femtojoule) electrons or γ -irradiation from a ^{60}Co source. Ferrous ion has been used to induce the redox decomposition of the macromolecular peroxy species at temperatures between 0 and 40°C . The amount of grafting was increased by using a reducing agent to decompose the peroxy species, thereby converting hydroxyl radical ($\dot{\text{O}}\text{H}$) to inactive hydroxide ion ($\ddot{\text{O}}\text{H}$). The γ -ray-induced vapor-phase grafting of methyl methacrylate (MMA) upon PP fiber was studied by Kamase and Hayakawa.⁷⁵ The rate of grafting increased linearly with irradiation time when the MMA vapor pressure was low but acceleration was observed for higher MMA vapor pressure.

The effect of PP γ -irradiation dose and temperature on the grafting of vinyl monomers was studied by Panchenkor *et al.*⁷⁶ The hydroperoxide content of PP at a 15 kGy dose showed two regions of irradiation temperature dependence — one above and the other below the T_g of PP. The authors found that use of an irradiation temperature below T_g formed movable radicals which were unable to react according to the classical peroxidation mechanism but which were trapped in the matrix and caused the independence of hydroperoxide groups at the irradiation temperature. The degree of grafting increased with irradiation

TABLE 2. Radiation grafting of *p*-nitrostyrene onto PP surface

Dose (kGy)	% grafting
5	11.9
10	17.5
15	27.0
20	30.2

time. The grafting⁷⁷ of styrene vapor was carried out on PP film with γ -irradiation under vacuum followed by exposure to oxygen at -78°C . The vacuum-irradiated sample showed more grafting than that irradiated in an oxygen atmosphere. In the former case, the trapped peroxy radicals had a higher mobility, leading to significant hydrogen abstraction by the peroxy radicals. This result means that the effective grafting radicals formed from the pre-irradiated PP on exposure to an oxygen atmosphere are carbon radicals, generated by hydrogen abstraction by peroxy-radicals.

Garnett *et al.*⁷⁸ have grafted *p*-nitrostyrene onto a PP surface by a γ -irradiation technique for the immobilization of a wide range of chemical and biochemical macromolecules. The grafting increased with the radiation dose up to a degree of 30% (Table 2). The nitro group in the poly(*p*-nitrostyrene-g-PP) after conversion to amino group (by the action of stannous chloride-hydrochloric acid) yielded a versatile polymer matrix which was utilized for the immobilization of biochemical macromolecules. A higher yield of γ -irradiation-induced styrene-grafted PP was obtained if the monomer solution in methanol was acidified with a mineral acid.⁷⁹

PP fiber was made dyeable with cationic dyes by grafting it with acrylonitrile/vinyl monomer containing a sulfonate group with γ -ray irradiation⁸⁰ under nitrogen. The fiber was readily dyed by basic dyes to produce bright light fast colours. PP was dipped in a solution of acrylamide and graft copolymerization was effected upon exposure of the PP to γ -rays for 16 hr giving a PP-acrylamide ($\sim 5\%$) graft copolymer.⁸¹ Burchill and Pinkerton⁸² had made a surface screen by grafting a polymerizable hydroxybenzophenone UV absorber onto a PP surface for effective photostabilization. Sivaram and Singh⁸³ have shown schematically the sequence of events that occur upon absorption of a photon of UV light. The γ -ray irradiation technique initiates graft copolymerization of the monomeric UV screener. If the radiation dose was kept at 0.5 Mrad or less, the mechanical properties of the PP were not affected seriously. Figure 3 shows variation in % grafting of methyl methacrylate onto PP at a 2 kGy dose.⁸⁴ The

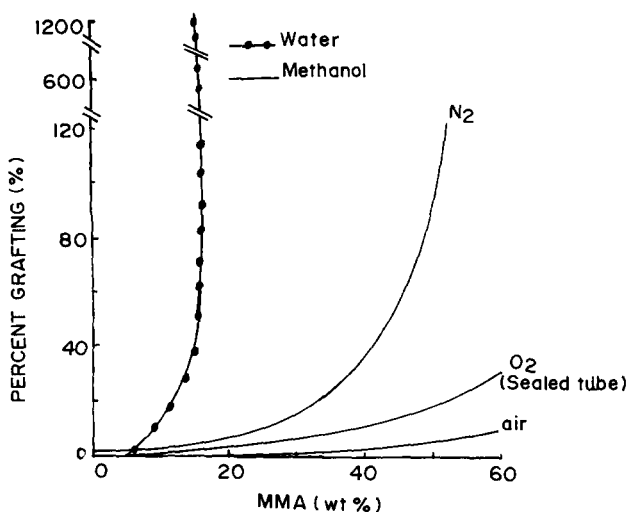


FIG. 3. Effect of methanol and water on MMA grafting onto PP film under γ -irradiated.

grafting was at a maximum in a nitrogen atmosphere because in air and oxygen, the oxygen is consumed by radicals produced by irradiation until polymerization is no longer significantly inhibited. The presence of acid in methanol enhances styrene grafting onto PP film, perhaps due to the Trommsdorff effect.⁷⁹ Water, however, also accelerates grafting even in the absence of acid (Fig. 3).

The post-irradiation grafting of styrene and methyl methacrylate onto *i*-PP was studied by Styrikovich *et al.*⁸⁵ The amount of grafting on a 50 μm annealed film increased fourfold when the annealing temperature was increased from 40 to 140°C. PP fibers were irradiated⁸⁶ with different doses of electron beam in the presence of acrylic acid and 2- and 4-vinylpyridine, giving grafted fibers with increased tenacity and dye uptake but decreased crystallinity. The graft polymerization of 4-vinylpyridine on PP in benzene was carried out by γ -rays in the absence of oxygen.⁸⁷ This graft was used as a support for transition metal catalysts. A colored PP film with good disperse dyeability and resistance to washing and dry cleaning was obtained⁸⁸ by γ -ray grafting with C_{7-11} alkyl methacrylates and 2-methyl-5-vinylpyridine at 30°C. PP films were grafted on the surface by ionizing radiation with 1-acryloyl-2-benzoylhydrazine and *N,N*-dimethylacrylamide to give grafts which form chelates with copper ions and develop fast colors.⁸⁹

1-(4-Diphenylphosphinophenyl)prop-1-ene and vinyldiphenyl phosphine⁹⁰ were grafted successfully onto a PP surface by γ -ray irradiation. The optimum

TABLE 3. Solvent and monomer uptake of PP fiber at 16 kGy dose

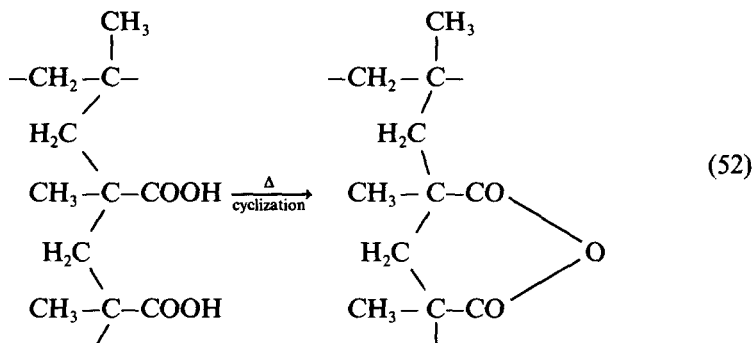
Temp. °C	Solvent uptake (%)	MA uptake (%)	MMA uptake (%)	AA uptake (%)
25 (No solvent swelling)	—	5.3	6.4	36.4
25	4.1	11.0	9.3	44.4
60	3.8	7.0	7.1	59.5
100	7.2	6.9	1.7	36.0

conditions for grafting were low dose rate and high total radiation in dimethyl sulfoxide in an inert atmosphere. Crown ether polymers were prepared⁹¹ by grafting unsaturated epoxides onto PP films by a γ -irradiation technique and condensing the product with an aminated crown ether. The γ -radiation induced grafting of acrylic acid onto PP film resulted in increased moisture sorption and dyeability.⁹² 4-Vinylpyridine was radiochemically grafted onto a PP surface to give graft copolymers having good ion-exchange capacity.⁹³ Alkylation of the graft gave a strongly basic anion exchanger, stable both in alkaline media and up to 300–350°C.

The radiation grafting of *N,N*-dimethylacrylamide onto PP film has been investigated by Ito *et al.*⁹⁴ They have confirmed in their studies that grafting may occur by mutual coupling, on pre-oxidized PP film or on trapped radicals, and in the amorphous regions of PP. In methyl acrylate (MA)/methyl methacrylate (MMA)/acrylic acid (AA) grafting onto PP surface by γ -irradiation, the presence of *n*-butanol increased the rate grafting of the monomers.⁹⁵ No specific trend was followed for monomer/solvent uptake at different temperatures, as is shown in Table 3.

The graft copolymerization of acrylonitrile onto PP fiber was studied by a preirradiation technique using γ -rays from a 78 terabecquerel ⁶⁰Co source as initiator.⁹⁵ The authors observed a maximum grafting rate in H₂O, a minimum rate in CH₃OH, and an intermediate rate in dioxane. Mukherji and Gupta⁹⁶ found in their studies that in γ -irradiation grafting of MMA onto PP fiber, MMA molecules were continuously diffusing into the fiber matrix. The same authors⁹⁷ evaluated the thermal behavior of PP-g-poly(methacrylic acid) fiber by TGA/DTA, DSC and limiting oxygen index measurement. The thermal properties of PP fiber were markedly improved by grafting MAA (methacrylic acid). The grafting occurred in the amorphous region only, without disturbing the crystalline part of the polymer. The improvement in thermal stability may be due to the abstraction of tertiary hydrogen atoms where grafting of PMMA chains occurs or due to a cyclization reaction involving PMAA moieties to form

a six-membered poly-methacrylic anhydride ring during heating:



An increase in moisture regain and dye uptake with increase in graft content was also observed. This was attributed to the hydrophilicity imparted to PP fibers by the polar carboxylic acid groups in the polymethacrylic acid grafts. Methyl methacrylate and/or methacrylic acid⁹⁸ were grafted onto PP fiber by γ -irradiation. An increase in water vapor sorption was observed upon grafting the fiber with both monomers. The kinetics of grafting of acrylic acid onto PP by irradiation of the PP followed by monomer exposure was studied by Mu and his coworkers.⁹⁹ The grafting rate increased with radiation dose, monomer concentration, time and temperature. Grafting onto the PP surface occurred by a radical mechanism and the activation energy was measured to be 107 kJ/mole. PP textile materials were ⁶⁰Co-irradiated in aqueous acrylic acid to obtain PP-g-acrylic acid¹⁰⁰ with 95% moisture regain capacity at 5% relative humidity.

Membranes for dialysis were prepared by grafting 2-hydroxyethyl methacrylate onto the surface of a PP thin film by ⁶⁰Co- γ -irradiation.¹⁰¹ The grafted membranes showed improved permeability towards urea and uric acid and their permeation coefficient was 5 to 19 times that of ungrafted PP membranes. Mehta and his coworkers¹⁰² grafted acrylonitrile onto *i*-PP by a preirradiation technique using a ⁶⁰Co- γ -ray source in the presence of ethanol, propanol, isopropanol, *n*-butanol, *tert*-butyl alcohol and *n*-pentanol. The graft-promoting effect of the alcohols decreased in the following order: *n*-pentanol > *n*-butanol > *tert*-butanol > *n*-propanol > ethanol > isopropanol. The alcohols of higher molecular weight appear, as expected, to interact with hydrophobic *i*-PP with greater facility resulting in enhanced grafting. The grafting occurs by a radical grafting mechanism. The authors have also studied the alcohol critical concentration beyond which the extent of grafting decreases, probably due to chain transfer.

Figure 4 shows that the IDT (initial decomposition temperature) values of *i*-PP, irradiated *i*-PP, *i*-PP-g-PAN and *i*-PP-g-PAAM are 205°C, 350°C, 270°C and 60°C respectively, and the final decomposition temperature ($T_{\max}$) values are 410°C, 460°C, 690°C and 535°C, respectively. Three-stage degradation was

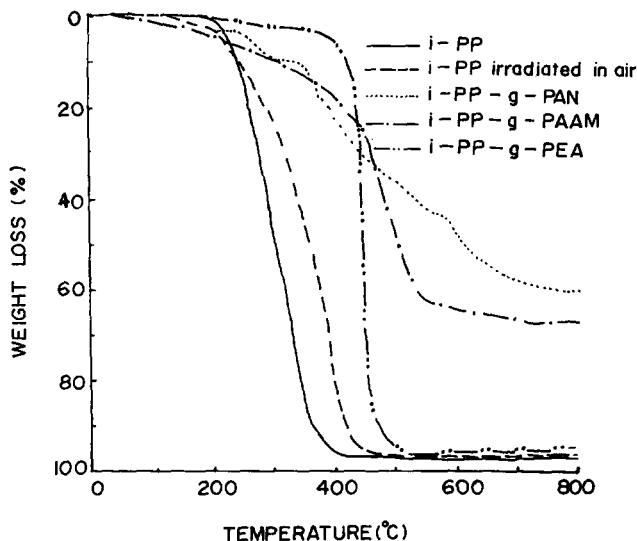


FIG. 4. Thermogravimetric analysis of *i*-PP, irradiated *i*-PP, *i*-PP-g-polyacrylonitrile (*i*-PP-g-PAN), *i*-PP-g-polyacrylamide (*i*-PP-g-PAAM) and A-PP-g-poly(ethyl acrylate) (A-PP-g-PEA).

observed in *i*-PP-g-PAN corresponding to maximum rates of degradation at 273°, 463° and 685°C. The weight loss starts at a low temperature (60°C) in *i*-PP-g-PAAM due to the presence of moisture. The residue left at 575°C is 77%, indicating that *i*-PP-g-PAAM possesses superior thermal properties in comparison with *i*-PP. It is also clear that acrylonitrile-grafted *i*-PP film shows better thermal properties than either *i*-PP, irradiated *i*-PP or *i*-PP-g-PAAM.

Tahes *et al.*¹⁰³ have directly grafted methacrylic acid by radiolysis onto PP film. The grafting was enhanced under vacuum irradiation in benzene as a diluent as compared with dimethylformamide, water, isopropanol, etc. They have confirmed that the grafting occurs by a diffusion controlled process. The surface of PP film and fiber was modified¹⁰⁴ by extruding and grafting it with polymerizable monomers by γ -irradiation.

4.3. Photografting

Polyolefin films were treated with aryl ketones and irradiated with ultraviolet light at ambient temperature in air to form hydroperoxides. These hydroperoxides were treated with a vinyl monomer in the presence of a reducing agent to give a graft copolymer of the polypropylene and vinyl compound. Polar free-radically polymerizable vinyl monomers¹⁰⁵ have been grafted on PP surfaces by carona discharge in the presence of oxygen. Before grafting, the PP

in the presence of benzophenone as a photosensitizer was studied by Kimura and Nakayama.¹¹⁴ The rate of photochemical grafting was found to depend on the rate of impregnation of these monomers in polymer films. The photochemically induced grafting kinetics of acryloyloxy substituted aromatic azo compounds e.g. 3-methyl-4-acryloyloxy-4'-(phenylazo)azobenzene (a model of an acrylated azodye) upon PP fiber was studied at 25°C by monochromatic irradiation.¹¹⁵ The effect of wavelength on the grafting was evaluated. The usual method of grafting involves two steps:



Yasuda *et al.*¹¹⁶ used a glow discharge to produce free-radicals on the PP surface, without any swelling or degradation of the polymer and grafted 2-vinylpyridine onto PP by this technique.

The effect of the use of 1,2-diphenyl-2,2-dimethoxyethane as photoinitiator upon the grafting of acryloyloxy-substituted azo dyes upon PP fiber was observed by Bellobono¹¹⁷ by employing crystalline dyes with polychromatic irradiation. Methacrylic acid/acrylamide mixture was photografted in the vapor phase onto a PP surface in the presence of benzophenone as a photosensitizer.¹¹⁸ The hydrophobic surface of PP film was hydrophilized by photografting and it has been established by ESCA and contact angle measurements that grafting occurred on the PP film surface. Vinyl polymers were thermally and photochemically grafted onto PP film (1 mm thick) in water using Bz₂O₂/benzophenone, as an initiator.¹¹⁹ The surface of thermally grafted samples was flat while photo-grafted surfaces appeared flat or grainy. The hydrophobic monomers (methyl methacrylate and styrene) provided the flat surfaces while the hydrophilic ones (acrylic acid/methacrylic acid and *N*-vinyl pyrrolidone) produced the grainy surfaces. The grafted chains in grainy surfaces were susceptible to photodecomposition.

1,1-Difluoroethylene was grafted¹²⁰ onto a PP film by a non-equilibrium plasma at atmospheric pressure in argon atmosphere. The grafting occurred via a radical mechanism. High degrees of grafting of vinyl monomers onto PP film were observed using benzophenone/anthraquinone sensitizer by a photo-grafting technique¹²¹ although the photoirradiation penetration into the polymer sample was considerably less than that of high energy radiation. The photo-grafting was limited mainly to the surface of the substrate due to the lower energy of light.

Hirotsu¹²² prepared water-ethanol permselective membranes by glow-discharge plasma grafting of acrylic acid/methacrylic acid and acrylamide onto a PP porous film. These membranes were used to separate azeotropic and isomeric mixtures. Recently this technique has been used for separation of water and alcohol from biomass products. Hirotsu *et al.*¹²³ have also grafted a hydrophilic

monomer (acrylamide) on the surface of PP film to make a permeable membrane which is used to separate water selectively from alcohol by pervaporation. Hirotsu and Nakajima¹²⁵ copolymerized acrylic acid with acrylamide onto a porous PP film which had been previously treated by a glow-discharge plasma. The membranes may be ionized by treating them with dilute sodium hydroxide. With different ratios of polyacrylate ions, these membranes were used for permseparation of water-ethanol mixtures.

The photostability of PP film photografted on the surface with a hindered amine light stabilizer (2,2,6,6-tetramethyl-4-piperidinylmethacrylate) was superior to that of PP film stabilized by addition of hindered amines without grafting.¹²⁴ The high effectiveness of surface photografting is due to the homogeneous grafting of HALS on the surface only, which is the part of the polymer exposed to weather. The grafting of acrylamide onto PP film surface¹²⁶ was carried out by immersion of argon-plasma-treated PP film in an aqueous acrylamide solution at 50°C. The grafting rates were found proportional to the discharge power and duration. Epailard and Brosse¹²⁷ have grafted multifunctional acrylate monomers onto PP after its surface was activated by cold plasma treatment. In this method grafting occurred because of a reaction of a macroradical (which had diffused to the surface of the substrate) with a reactive species created on the PP surface. A light stabilizer (2,2,6,6-tetramethyl-4-piperidinyl methacrylate) was photochemically grafted onto PP film¹²⁸ in the presence of benzophenone. The benzophenone is a free-radical initiator and the grafted film showed improved photostability and wettability.

The UV-radiation-induced¹²⁹ grafting of PP film with styrene in methanol was significantly enhanced in the presence of sulphuric acid, lithium salts, urea and polyfunctional ureas. The enhancement with polyfunctional monomers was due to branching of the grafted polystyrene chains, while with additives it was because of changes in the partitioning of styrene monomer between the solution and the substrate. The combination of both resulted in a synergistic improvement in grafting efficiency.

Toshio and his coworkers¹³⁰ grafted *N,N*-dimethylacrylamide onto a microporous PP membrane by plasma graft copolymerization. In their studies they found that even short exposure to argon and helium plasma (< 10 s) resulted in very fast graft polymerization. Both the peroxy and alkyl radicals could initiate the grafting and their activation energies were 59.4 and 33.1 kJ/mole, respectively. Thermal and X-ray analysis revealed that grafting occurred not only at the inner and outer surfaces but also in the bulk of the substrate.

Allmer and his coworkers^{131,132} have attached epoxy functional groups to a PP surface by photografting it with glycidyl acrylate and/or glycidyl methacrylate. They have attached 4-amino-2,2,6,6-tetramethyl-piperidine, 2,4-dihydroxybenzophenone and phenyl-4-aminosalicylate to this reactive surface. They then studied the UV stability of the grafted films by exposing them to UV irradiation in air. The phenyl salicylate and dihydroxybenzophenone act as UV absorbers

and screens,¹³³ while the hindered piperidine acts as a radical scavenger and hydroperoxide decomposer.¹³⁴ The PP-g-phenyl salicylate is the least stable, while PP-g-hindered piperidine is the most stable film. This result may be due to the susceptibility of phenyl salicylate to a photo-Fries rearrangement where the main product is 2,2'-dihydroxybenzophenone. Lucki *et al.*¹³⁵ have grafted polymerizable monomers containing UV stabilizer groups as 2-(2-hydroxyphenyl)-2H-benzotriazole onto the surface of a PP film in the presence of benzophenone as a photoinitiator using 254 nm UV irradiation. The grafting occurred on the degraded sites of the PP molecules and it was found that the photografted films have much higher resistance towards photodegradation than the ungrafted films.

Acrylamide¹³⁶ was photochemically grafted onto a PP surface by exposing a mixture of PP, acrylamide and traces of benzophenone in acetone to UV rays. The presoaking of PP in acetone facilitates an increased rate of photografting. The dye absorption capacity and adhesion of the grafted film increased three to four times compared with the ungrafted one.

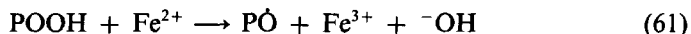
4.4. Miscellaneous grafting

PP fiber can be made easily dyeable and hygroscopic by grafting poly(acrylic acid) onto its surface. The grafting was accomplished¹³⁷ by hydroperoxide formation on the PP fibers by atmospheric oxidation. In the reaction medium ferrous sulphate, a reducing agent, is added to suppress homopolymerization because it decomposes the peroxidized species and scavenges the free radicals. Mikulasova¹³⁸ has grafted acrylonitrile and styrene on partially oxidized pulverized PP by a redox system. A higher rate of grafting was reached in the mixed system than with individual monomers.

Narębska and Bukowski¹⁴¹ have prepared a PP graft by immersing a PP film in an aqueous solution of acrylic acid containing ferrous sulphate as activator. The hydroperoxide groups in the PP film produce the macroperoxy radicals via thermal decomposition or by reacting with ferrous ions.



or



The macroradical (P $\dot{\text{O}}$) initiates the grafting with acrylic acid:



The oxidized *i*-PP powder was grafted at 30°C in an aqueous emulsion of water soluble vinyl monomer.¹⁴² The decomposition of macroperoxy groups initiated the graft copolymerization by a radical mechanism in the presence of ferrous chelate.

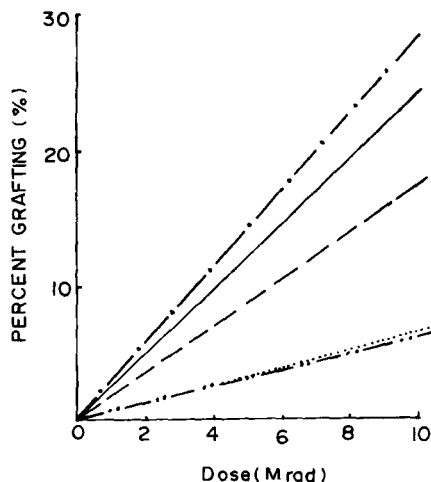


FIG. 5. Grafting of ethyl vinyl ether onto PP at various doses rates: 49.4 rad/s (—), 70.6 rad/s (---), 36.7 rad/s: in air (— — —), *n*-pentane (— · —) and neopentane (····).

Ethyl vinyl ether has been grafted onto PP film under “super-dry” conditions. Kabanov *et al.*¹³⁹ indicated that the process followed a radiation-induced ionic mechanism as the grafting did not take place in the presence of traces of water. The percentage of grafting (Fig. 5) increased linearly with dose up to 30% grafting under the best conditions at 10 Mrad.

A high yield of acrylonitrile-styrene grafting onto an atactic PP surface was obtained¹⁴⁰ by chlorinating the PP and then grafting it using TiCl_3 initiator by an ionic (or coordinative) grafting technique in DMF solvent.

5. FUTURE TRENDS

Grafting is a common method of modifying polymer properties. The main chain and the branch chains are usually thermodynamically incompatible. Therefore most graft copolymers can be classified as multiphase polymers in the solid state, analogous to polymer blends, blocks and interpenetrating polymer network.^{143,144} Although there is similarity in processing and fabrication of graft copolymers and many other polymeric materials, the graft copolymers have a greater sensitivity towards processing and fabrication due to their multi-component nature. Grafting improves adhesion, tensile strength, abrasion resistance, dyeing and dye retention capacity of the copolymer. Similarly crease resistance and crease retention of cotton are increased by grafting it with vinyl monomers. Grafting also improves compatibility and thermal and photochemical stability. The grafting of UV stabilizers on polymer surfaces is most useful as the stabilizer suppresses the formation of initiating species on the

surface, shields the rest of sample from UV radiation, and is firmly attached to the surface, because of which the risk of stabilizer loss by migration and evaporation is minimized.

In recent times, *i*-PP has become one of the largest and fastest growing plastics due to its versatility, wide applications and low cost. The inertness of this polymer towards chemicals hinders industrial applications such as dyeing, printing and adhesion, etc. In order to improve these and other properties of *i*-PP, its modification became necessary. The free-radical polymerization methods are the oldest, most widely used and inexpensive procedures for the synthesis of graft copolymers because they are relatively simple. It is likely that the utility of graft copolymers will continue to expand as more is learned about their synthesis and unique properties.

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